Chapter test

31 Mechanisms of evolution and speciation

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following is ***not*** part of Darwin's theory of natural selection?

   A. Offspring inherits half of its genes from each of its parents.
   B. Variations exist between individuals of the same species.
   C. Organisms produce more offspring than they can survive.
   D. Population size remains relatively constant and cannot get bigger indefinitely. ☐

2. Which of the following causes the struggle for survival in a population?

   A. accumulation of variations
   B. over-production of offspring
   C. unlimited resources
   D. occurrence of mutation ☐

3. Which factor(s) is/are essential for evolution to occur within a population?

   (1) change in environmental conditions
   (2) variation in the population
   (3) natural selection

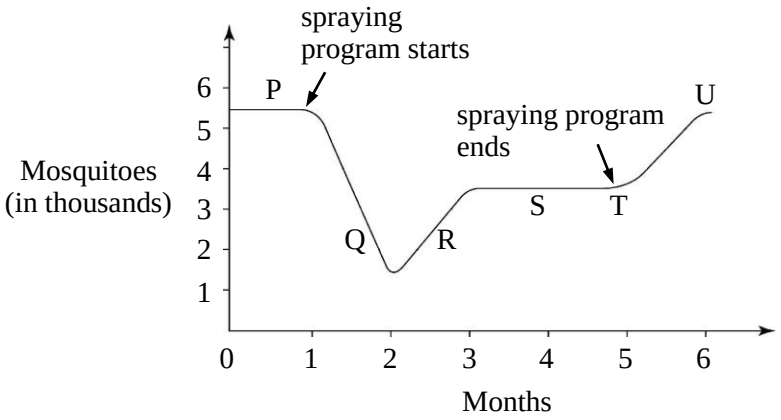
   A. (1) only
   B. (1) and (2) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

4. The modern horse is much larger than its ancestors. Which of the following is the best explanation for this change?

   A. The ancestors of the modern horse had to stretch for young leaves on the trees. Thus, they developed longer legs and necks.
   B. Mutation occurred in the past, leading to the formation of larger horses.
   C. The ancestors of the modern horse had to run very fast to escape from predators. Thus, they developed stronger muscles and longer legs.
   D. Larger horses are more adaptable to the environment than the smaller ones. Thus, they are able to survive and breed. ☐

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Directions: Questions 5 and 6 refer to the information below. A tropical island was inhabited by mosquitoes. DDT was sprayed over a period of several months. The change in population size of the mosquitoes was shown below.



★ 5. Which of the following statements about the graph is correct?

A. DDT is ineffective in killing the mosquitoes.
B. The population of the mosquitoes decreases throughout the six months.
C. The population of the mosquitoes decreases during the spraying program.
D. Only female mosquitoes are killed by DDT. ☐

★ 6. The difference in the genetic make-up of the mosquitoes is the greatest between points

A. P and S.
B. P and U.
C. Q and S.
D. R and U. ☐

Directions: Questions 7 and 8 refer to the experiment below.

A large number of peppered moths were captured and marked with paint for identification. 504 light moths and 497 dark moths were released. After several days, 52 light moths and 30 dark moths were recaptured.

★ 7. Which of the following statements about the experiment is correct?

A. Equal numbers of moths should be released in order to get a conclusive result.
B. Larger number of moths should be recaptured in order to give a conclusive result.
C. The moths were released in the industrial area.
D. The moths were released in the countryside. ☐

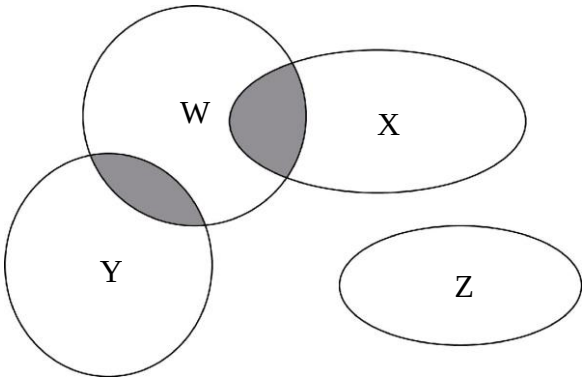
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★ 8. Which of the following combinations is correct?

	Position of marking	*Reason*
A.	on the upper side	easier to identify the recaptured moths
B.	on the upper side	easier to mark the moths on this side
C.	on the under side	the mark would not cause bias during recapture
D.	on the under side	paint wears off on the upper side more easily

☐

Directions: Questions 9 and 10 refer to the diagram below which shows four populations. Interbreeding takes place in the shaded areas to produce fertile offspring.



★ 9. How many species are present?

A. 1
B. 2
C. 3
D. 4

☐

★★ 10. How many species would be present if population W had disappeared for over a hundred years?

A. 1
B. 2
C. 3
D. 4

☐

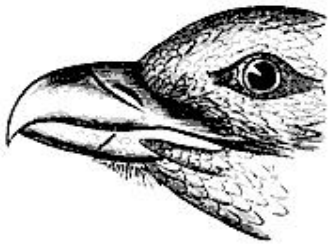
Question	1	2	3	4	5	6	7	8	9	10
Related section	31.1	31.1	31.1	31.1	31.1	31.1	31.1	31.1	31.2	31.2

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B. Problem solving (22 marks)

11. The Galapagos Islands are about 1000 km from the mainland of South America. There is one species of finch living on the mainland, but there are many species found on the islands. On one of the islands, there are two finch species. Their heads are shown below.

A





(a) Suggest the diets of the above finches. Explain your answer. (2 marks)

(b) State ***two*** ways in which competition between species A and B is reduced. (2 marks)

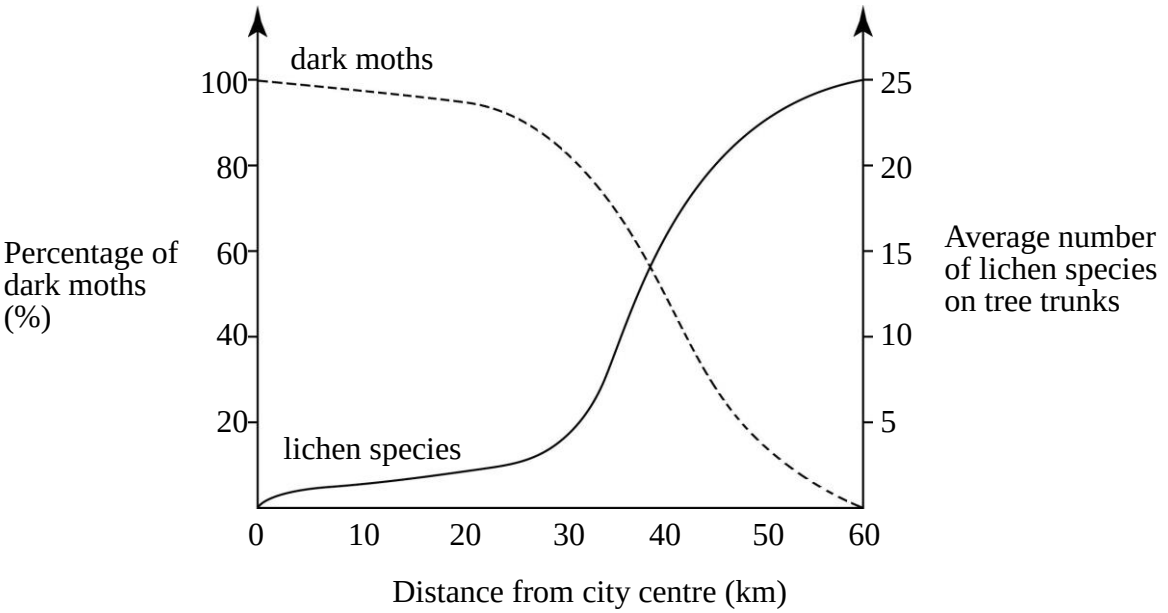
(c) Darwin suggested that the two finch species were evolved from a finch species which had arrived from the mainland. The original finch probably had a thick beak. Suggest how the thin beak species could have evolved. (3 marks)

(d) Suggest ***two*** reasons why there are more species of finches on the island than on the mainland of South America. (2 marks)

(Total: 9 marks)

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★ 12. There are two types of peppered moths in Britain: light moths and dark moths. They rest on tree trunks which may be covered by lichens. The graph below shows the results from a survey of lichens and the dark moths done in the vicinity of an industrial city in the 1950s.



(a) Describe the relationship between the distribution of lichens and dark moths. (1 mark)

(b) Explain the distribution curve of the dark moths. (3 marks)

(c) If the distribution curve of the light moths is drawn on the graph, what would be the form of the curve? (1 mark)

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(d) At the centre of the city, the percentage of dark moths decreased from 95% to 81% after the Clean Air Act in 1956. Explain why. (2 marks)

(Total: 7 marks)

★ 13. The table below shows the events of rabbit population control in Australia.

Year	Event
1859	Europeans released 24 rabbits in Australia. There were no natural predators of rabbits in Australia.
1950	The rabbit population had increased to over 60 millions, causing serious damage to crops.
1954	A virus, which can kill rabbits, was introduced to control the rabbit population.
1960	The rabbit population had decreased by 90%.
2010	The rabbit population had increased back to over 30 millions.

Using the concept of natural selection, explain the changes in the rabbit population in Australia. (6 marks)

(Total: 6 marks)

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C. Exam challenge (8 marks)

★★ 14. The brown rat (*Rattus norvegicus*) originally lived in woodlands but today most brown rats live in urban cities.

(a) Suggest ***two*** reasons why most brown rats now live in urban cities. (2 marks)

(b) Rodenticides are chemicals used to kill rats. Many populations of rats are becoming resistant to a type of rodenticide (W). Scientists captured brown rats from two different populations (P and Q) and gave the rodenticide W to each group of brown rats.

- Population P was from a city where the rodenticide W had been used for five years.
- Population Q was from a different city 20 km away, where the rodenticide W had not been used previously.

Scientists found that resistance to rodenticide W has occurred in one of the populations.

(i) Suggest which population has evolved resistance to rodenticide W, and explain the mechanism of evolution by natural selection. (5 marks)

(ii) Suggest why the other population has not evolved resistance to rodenticide W. (1 mark)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	A	2.	B	3.	D	4.	D	5.	C
6.	B	7.	D	8.	C	9.	B	10.	C

(1 mark each)

B. Problem solving (20 marks)

11. (a) Species A has a large and thick beak to crush large seeds, nuts or insects. (1)
Species B has a slender beak to catch small insects from the holes on the bark of trees. (1)

(b) They feed on different types of food. (1)
They live in different habitats. (1)

(c) The original finches that arrived at the island had thick beak, but there was not enough food for all the birds. (1)
Finches with thin beaks were able to eat small insects from the holes on the bark of trees. (1)
These finches survived to breed and passed on their traits to their offspring. (1)

(d) More niches on the islands /
Wider range of habitats on the islands /
Less competition on the islands (Any ***two***, 2)

(Total: 9 marks)

12. (a) The percentage of dark moths decreases as the number of lichen species increases. (1)

(b) At the city centre, pollutants released from factories caused blackening of the tree trunks.
The lichens were killed by these pollutants. (1)
Dark moths were best camouflaged on dark tree trunks and can avoid predation. (1)
More lichens grew on tree trunks outside the city. The dark moths were no longer camouflaged and were therefore easily spotted by predators. (1)

(c) The curve for the light moths would be similar to that of the lichens. (1)

(d) The polluted areas were becoming cleaner in response to the Clean Air Act. More lichens can grow in these areas. (1)
The dark moths were no longer camouflaged and were easily spotted by predators. (1)

(Total: 7 marks)

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13. There were no predators for rabbits at the beginning. There were sufficient food / space / suitable climate for the growth of the rabbits. (1)
Due to low competition, the rabbits reproduced rapidly from 1859 to 1954. (1)
Due to genetic variations / mutations, some rabbits were resistant to the virus introduced in 1954. (1)
The resistant rabbits had selective advantage over non-resistant rabbits. (1)
Most of the non-resistant rabbits were killed by the virus, led to population drop in 1960. The survived rabbits face less competition for resources and they can reproduce quickly. (1)
The virus-resistant gene was passed to the next generations. After many generations, most rabbits are resistant to the virus and the population increased again. (1)

(Total: 6 marks)

C. Exam challenge (8 marks)

14. (a) Loss of original habitat due to deforestation /
Increased urbanization /
Urban cities provide more food supply / shelter
(Or other reasonable answers) (Any ***two***, 2)

(b) (i) Population P (1)
Some rats in the population are resistant to the rodenticide W due to random mutations / genetic variations. (1)
The resistant rats are not killed by the rodenticide, while the non-resistant rats are killed. (1)
The resistant rats can obtain more resources and they have a higher chance to survive and reproduce. (1)
They have a higher chance to produce offspring which are resistant to the rodenticide W. Over many generations, the proportion of resistant rats increases in the population. (1)

(ii) No selection pressure for the resistant rats in population Q /
Mutation which produces rodenticide resistance had not occurred in population Q /
No interbreeding / gene flow between populations P and Q
(Or other reasonable answers) (Any ***one***, 1)

(Total: 8 marks)

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